Recap: Function → Assignment of a unique value for each element in the domain.

Algebra on Space of Functions ($\mathcal{F}(\mathbb{R})$):

Addition, multiplication, scalar multiplication, division (Pointwise.)

Bounded functions / Unbounded functions

Ex: i) Let $f: [1, \infty) \to \mathbb{R}$ with $f(x) = 1/x$.

This function, is bounded above by 1, and below by 0.

A new function $g: (0, 1] \to \mathbb{R}$ given $g(x) = 1/x$ is not equal to $f(x)$ since $[1, \infty) \neq (0, 1]$. Also g is bounded below by 1 but now not bounded from above.

ii) Let $f: [0, 3] \to \mathbb{R}$ given by $f(x) = x^2$. This function is bounded above by 9 and below by 0.

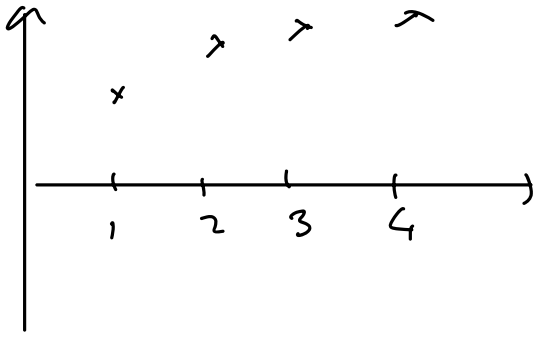
Theorem 2.10 (Bounded Functions)

Let $f: S \to \mathbb{R}$. Then f is bounded iff then exists $K \in \mathbb{R}$ such that $|f(x)| \leq K \quad \forall x \in S$.

Continuous Functions

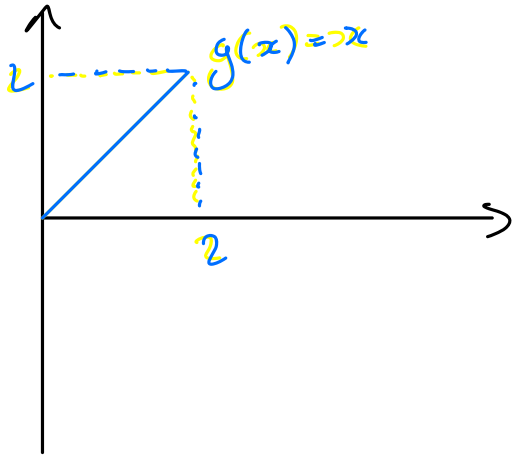
Recall sequences are defined as $a: \mathbb{N} \to \mathbb{R}$.

These are real functions, and their graphs look like a bunch of discrete points.



What about if we define a function with domain some interval of $\mathbb{R}$? In this case, since $\mathbb{R}$ is complete, and an ordered field, the nature of graphs of functions on this domain can be quite different.

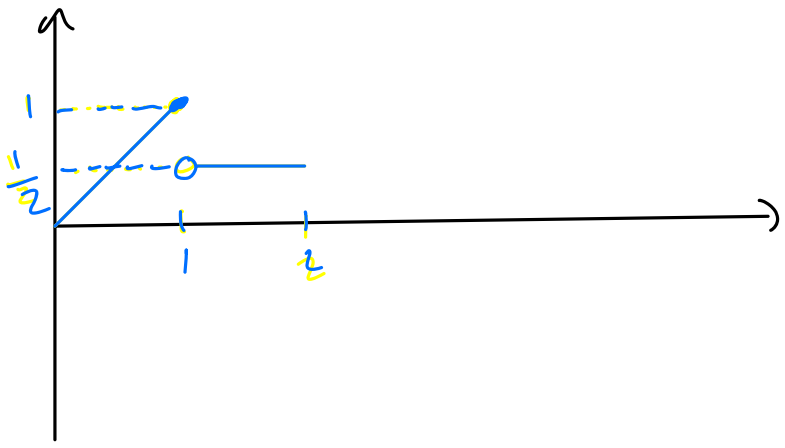
For example, $g: [0,2] \to \mathbb{R}$ given $g(x) = x$. Then its graph looks like



Consider the function $f: [0,2] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x & : x \in [0,1] \\ 1/2 & : x \in (1,2] \end{cases}.$$

The graph of this function looks like



Intuitively, we think of a continuous function as one we can draw without taking our pen off the paper. If we formalise this, we get the definition of what is known as a Darboux function.

Defn: Darboux Function

Let $f: D \to \mathbb{R}$. Then the function is called Darboux if for all $[a,b] \subset D$, and $\alpha \in \mathbb{R}$, with $f(a) < \alpha < f(b)$, or $f(b) < \alpha < f(a)$,

there exists a $c \in (a,b)$ with $f(c) = \alpha$.

(cts)

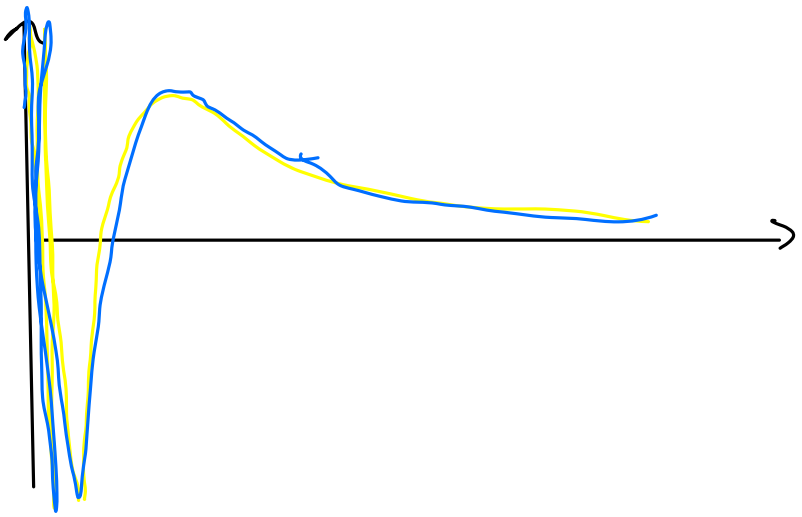
1) One property we would expect for continuous functions, is that they are bounded on closed intervals (compact sets).

For Darboux functions, this is not necessarily true.

For example, $f : [0,1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \frac{1}{x} \sin\left(\frac{1}{x}\right) & : x \in (0,1] \\ 0 & : x = 0 \end{cases}.$$

The graph looks like



This is Darboux, but unbounded.

2) We would also like our definition of continuity to be preserved under the algebraic operations defined

on $\mathcal{F}(\mathbb{R})$. This is not the case for Darboux functions.

For example, $g: [0,1] \to \mathbb{R}$ and $h: [0,1] \to \mathbb{R}$

with $g(x) = \begin{cases} \sin^2(1/x) : x \in (0,1] \\ 0 : x = 0 \end{cases}$ and

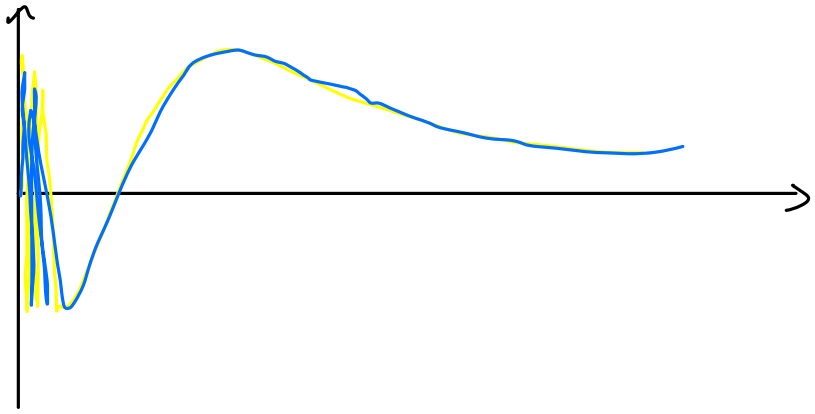
$h(x) = \begin{cases} \cos^2(1/x) : x \in (0,1] \\ 0 : x = 0 \end{cases}$.

Then $g+h : [0,1] \to \mathbb{R}$ is given by

$$(g+h)(x) = \begin{cases} 1 : x \in (0,1] \\ 0 : x = 0 \end{cases}.$$

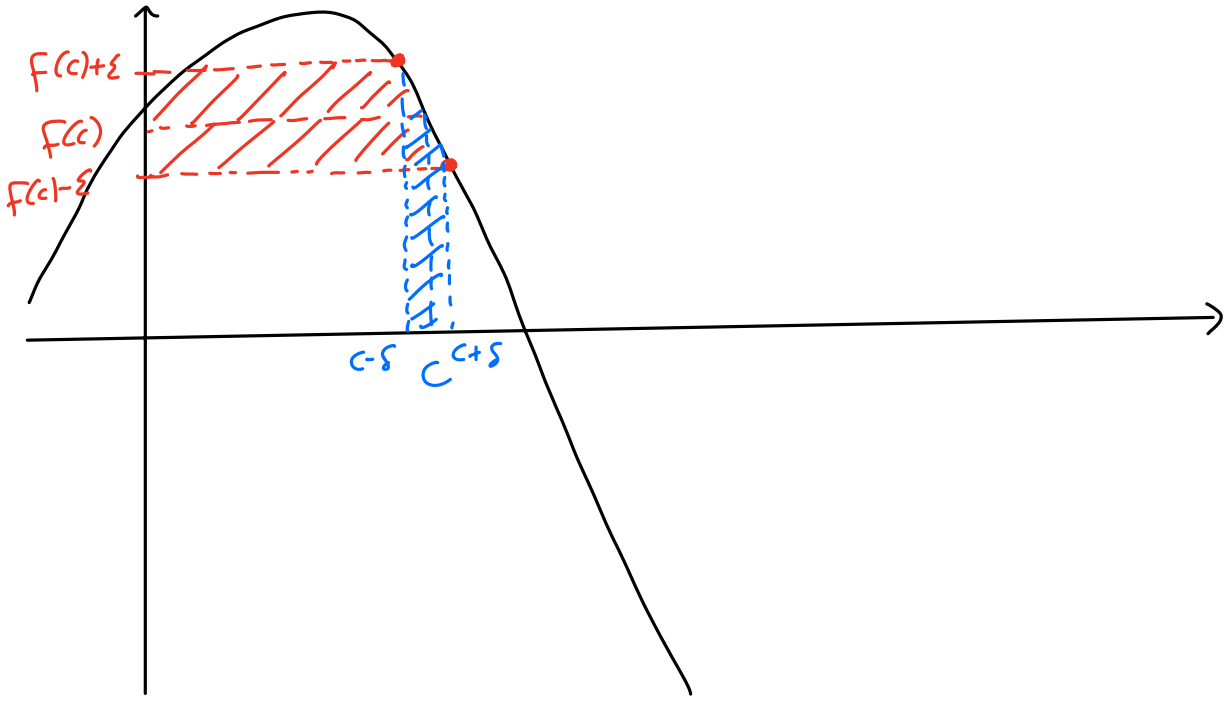
3) The final property we would like for cts functions is for small changes in the domain to lead to small changes in the co-domain. This is not true for Darboux functions. Consider, the <u>topological sine curve</u> given by $f: [0,1] \to \mathbb{R}$, $f(x) = \begin{cases} \sin(1/x) : x \in (0,1] \\ 0 : x = 0 \end{cases}$.

The graph looks like



We see that any small change from zero, leads to large changes in the value of the function

We want our definition of continuity to be that small changes around a value c, in the domain of the function, leads to arbitrarily small changes around $f(c)$, in the co-domain of the function. That is, squeezing the region around c, leads to squeezing the region around $f(c)$.



Defn: (Topological Continuity)

A real function $f: D \to \mathbb{R}$ is <u>continuous at a point $c \in D$</u> if for every ε-neighbourhood of $f(c)$, ($V_\varepsilon(f(c))$) there exists a δ-neighbourhood of c ($V_\delta(c)$), such that $f(x) \in V_\varepsilon(f(c))$ whenever $x \in V_\delta(c)$ and $x \in D$.

Defn: (Metric Continuity)

A real function $f: D \to \mathbb{R}$ is <u>continuous at a point $c \in D$</u> if for all $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon$$

whenever $|x - c| < \delta$ and $x \in D$.

In general, the value of δ will be a function of ε.

Ex: Let $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = 10x$.

Then f is cts at $c = 2$.

Scratch Work:

$$|f(x) - f(2)| = |10x - 20|$$

$$= 10|x - 2|.$$

Well $|x-2| < \varepsilon/10$. Then $|f(x) - f(2)| < \varepsilon$.

Proof: Given $\varepsilon > 0$, define $\delta = \varepsilon/10$. It follows that if $|x - 2| < \delta$ we have

$$|10x - 20| < 10|x - 2|$$

$$< 10\delta$$

$$= \varepsilon. \quad \square$$

Ex: Let $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$.

Then f is cts at $c = 3$.

Scratch Work:

$$|f(x) - f(3)| = |x^2 - 9|$$

$$= |x-3||x+3|$$

Say $|x-3| < 1 \iff -1 < x-3 < 1$

$$\iff 2 < x < 4$$

$$\iff 5 < x+3 < 7.$$

$$\Rightarrow |x+3| < 7$$

Then, $|f(x) - f(3)| < |x+3||x-3|$

$$< 7|x-3| \quad (\delta = \varepsilon/7)$$

$$< \varepsilon.$$

Proof: Given $\varepsilon > 0$, define $\delta = \min\left(1, \varepsilon/7\right)$.

Then if $|x-3| < \delta \leq 1$ it follows that

$$\begin{aligned} |f(x)-f(3)| &= |x^2 - 9| \\ &= |x+3||x-3| \\ &\leq 7|x-3| \\ &< \frac{7\varepsilon}{7} \\ &= \varepsilon. \end{aligned}$$

□

Defn: Let $f: D \to \mathbb{R}$. Then the function f is said to be cts on the interval I if it's cts for each point $c \in I$.

Properties of Continuous Functions

Theorem: Algebraic Properties of Cts Functions

Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ both be cts at the point $c \in D$. Then

i) $f+g: D \to \mathbb{R}$ is cts at $c \in D$

ii) $fg: D \to \mathbb{R}$ is " at $c \in D$

iii) $f/g: D \to \mathbb{R}$ " " " $c \in D$, $g(x) \neq 0$ $\forall x \in D$

iv) $\lambda f: D \to \mathbb{R}$ " " " $c \in D$.

Proof: For every $\varepsilon > 0$, there exists $\delta_1 > 0$ s.t

$$|f(x) - f(c)| < \varepsilon/2$$

whenever $|x - c| < \delta_1$ and $x \in D$. Similarly, $\exists\, \delta_2 > 0$ s.t

$$|g(x) - g(c)| < \varepsilon/2$$

whenever $|x - c| < \delta_2$ and $x \in D$. Define $\delta = \min(\delta_1, \delta_2)$. Then if $|x - c| < \delta$ and $x \in D$

we have that

$$|f(x)+g(x)-f(c)-g(c)|$$

$$\overset{\text{T.I.}}{\leq} |f(x)-f(c)| + |g(x)-g(c)|$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$< \varepsilon. \quad \square.$$